INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

BOLT, SELF-RETAINING, POSITIVE LOCKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers positive locking self-retaining bolts.

1.2 Classification. Bolts will be of the following types and classes, as specified (see 6.2):

Types

Type I - Head end actuated.

Type II - Thread end actuated (solid shank).

Classes

Class 1 - Corrosion resistant steel bolts, magnetic (PH13-8Mo).

Class 2 - Corrosion resistant steel bolts, nonmagnetic (A286).

Class 3 - Titanium bolts (6Al-4V).

Class 4 - Nickel Alloy (718).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, SESD (Code 53), Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-P-35	Passivation Treatments for Corrosion-Resisting Steel
QQ-S-763	Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting
QQ-W-390	Wire, Nickel-Chromium-Iron Alloy
PPP-H-1581	Hardware (Fasteners and Related Items), Packaging and Packing for Shipment and Storage of

MILITARY

MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-H-6875	Heat Treatment of Steels (Aircraft Practice) Process for
MIL-S-8879	Screw Threads, Controlled Radius Root With Increased Minor Diameter; General Specification for
MIL-L-46010	Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting
MIL-H-81200	Heat Treatment of Titanium and Titanium Alloys

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
AN310	Nut, Plain, Castellated, Airframe
AN320	Nut, Plain, Castellated, Shear
MIL-STD-810	Environmental Test Methods
MIL-STD-1312	Fasteners, Test Methods
MIL-STD-1949	Inspection, Magnetic Particle
MS3369	Bolts, Self-Retaining, Positive Locking, CRES 90 KSI Fsu, 63 KSI Ftu, Hexagon Slotted Head, 450°F and 650°F
MIL-STD-6866	Inspection, Liquid Penetrant

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STANDARDS (continued)

MILITARY (continued)

MS17825	Nut, Self-Locking, Castellated, Hexagon, 250°F Non-Metallic Insert
MS17826	Nut, Self-Locking, Castellated, Hexagon, 250°F, Thin Non-Metallic Insert
MS20002	Washer, Countersunk and Plain, High Strength
MS21125	Bolts, Self-Retaining, Positive Locking, CRES 90 KSI Fsu, 63 KSI Ftu, Pan Head, 450°F and 650°F
MS21130	Bolts, Self-Retaining, Positive Locking, CRES 90 KSI Fsu, 63 KSI Ftu, 100° Flush Head, 450°F and 650°F
MS21133	Nut, Self Locking, Steel, 180 KSI Ftu, 450 Degree F, Flanged, MS33787 Wrenching Element
MS21224	Nut, Self-Locking, Castellated, Hexagon, Counterbored, Assembled Washer, 250°F, Non-Metallic Insert (for Self-Retaining Bolts)
MS21244	Nut, Castellated, Hexagon, Counterbored Assembled Washer, 450°F (for Self-Retaining Bolts)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building No. 4, Section D, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.

AEROSPACE MATERIAL SPECIFICATIONS

AMS4928	Titanium Alloy Bars and Forgings (6Al-4V)
AMS4967	Titanium Alloy Bars and Forgings (6Al-4V) Annealed, Heat Treatable
AMS5629	Steel Bars, Forgings, Tubing, and Rings, Corrosion Resistant 13CR - 8Ni - 2.3Mo - 1.1Al Vacuum Induction Plus Consumable Electrode Melted

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AEROSPACE MATERIAL SPECIFICATIONS (continued)

AMS5662	Alloy Bars, Forgings, and Rings Corrosion and Heat Resistant 52.5Ni - 19Cr - 3.0 Mo - 5.1(Cb+Ta) - 0.90 Ti - 0.50 Al - 18Fe Consumable Electrode or Vacuum Induction Melted 1775°F (970°C) Solution Heat Treated
AMS5663	Alloy Bars, Forgings, and Rings, Corrosion and Heat Resistant 52.5Ni - 19Cr - 3.0Mo - 5.1(Cb+Ta) - 0.90 Ti - 0.50 Al - 19Fe Consumable Electrode or Vacuum Induction Melted 1750°F (955°C) Solution and Precipitation Heat Treated
AMS5678	Steel Wire, Corrosion Resistant 17Cr - 7.1Ni - 1.1Al, Precipitation Hardenable, Spring Temper
AMS5699	Alloy Wire, Corrosion and Heat Resistant, Nickel Base 15.5Cr - 7Fe - 2.3Ti - I (Cb-Ta) - 0.7Al
AMS5737	Steel Bars, Forgings and Tubing, Corrosion and Heat Resistant 15Cr - 26Ni - 1.3Mo - 2.1Ti - 0.30V Consumable Electrode Melted 1650°F (898.9°C) Solution Treated and Precipitation Heat Treated
AMS5833	Alloy Wire, Corrosion and Heat Resistant 20Cr - 15Ni - 40Co - 7.0Mo - 0.06Be - 16Fe Solution Heat Treated and Cold Drawn
AMS5834	Wire, Corrosion and Heat Resistant 40Co - 20Cr - 15Ni - 7.0Mo - 0.06Be - 16Fe Solution Heat Treated, Cold Drawn, and Aged

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI B46.1-85	Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, NY 10016.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E8	Standard Test Methods of Tension Testing of Metallic Materials

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL AEROSPACE STANDARDS

NAS 618 Fastener - Recommended Shank, Hole and Head-to-Shank Fillet Radius Limits for

(Application for copies should be addressed to the National Standards Association, Inc., 1725 DeSales Street N.W., Washington, DC 20036.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 MS sheets. The individual item requirements shall be as specified herein and in accordance with the applicable MS sheets. In the event of any conflict between the requirements of this specification and the MS sheets, the latter shall govern.

3.2 Qualification. The self-retaining bolts furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.3 and 6.3).

3.3 Materials.

3.3.1 Identification. The manufacturer shall make material identification tests of the metal stock used to fabricate the bolts. A test report indicating that the identification tests were made shall be made available upon the request of the acquiring activity or as directed by the contracting officer (see 4.3.2).

3.3.2 Class 1 bolts. Bolt bodies and plungers shall be fabricated from corrosion-resistant steel conforming to the chemistry of AMS5629 (PH13-8MO).

3.3.2.1 Class 1 locking balls or elements. Locking balls or elements shall be fabricated from material as specified in QQ-S-763 Class 440C, AMS5629 (PH13-8Mo) or AMS5663 (718).

3.3.2.2 Class 1 springs. Springs shall be fabricated from 17-7PH as specified in AMS5678; or from elgiloy wire in accordance with AMS5833 or AMS5834.

3.3.3 Class 2 bolts. Bolt bodies and plungers shall be fabricated from corrosion-resistant steel conforming to the chemistry of AMS5737 (A286).

3.3.3.1 Class 2 locking balls or elements. Locking balls or elements shall be fabricated from material as specified in AMS5737(A286) dr AMS5663 (718).

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3.3.3.2 Class 2 springs. Springs shall be fabricated from material as specified in QQ-W-390, Temper C or AMS5699, or from elgiloy wire in accordance with AMS5833 or AMS5834.

3.3.4 Class 3 bolts. Bolt bodies shall be fabricated from titanium conforming to AMS4928 (6Al-4V) or AMS4967 (6Al-4V).

3.3.4.1 Class 3 plungers. Plungers shall be fabricated from titanium conforming to AMS4928 (6Al-4V) or AMS4967 (6Al-4V), AMS5629 (PH13-8Mo) or AMS5663 (718).

3.3.4.2 Class 3 locking balls or elements. Locking balls or elements shall be fabricated from titanium alloy material as specified in AMS4928, from magnetic material as specified in QQ-S-763, Class 440C, AMS5663 (718) or AMS5629 (PH13-8Mo).

3.3.4.3 Class 3 springs. Springs shall be fabricated from 17-7PH as specified in AMS5678; or from elgiloy wire in accordance with AMS5833 or AMS5834.

3.3.5 Class 4 bolts. Bolt bodies shall be fabricated from nickel alloy 718 per AMS5662 or AMS5663.

3.3.5.1 Class 4 plunger. Plungers shall be fabricated from nickel alloy AMS5662, AMS5663 (718) or AMS5629 (PH13-8Mo).

3.3.5.2 Class 4 locking balls or elements. Locking balls or elements shall be fabricated from material specified in QQ-S-763 Class 440C, AMS5629 (PH13-8Mo) or AMS5663 (718).

3.3.5.3 Class 4 springs. Springs shall be fabricated from 17-7PH as specified in AMS5678; or from elgiloy wire in accordance with AMS5833 or AMS5834.

3.3.6 Spring retaining plug. The spring retaining plug, if used, shall be manufactured from the same material as the bolt. Heat treatment is optional.

3.4 Passivation. Corrosion-resistant steel shall be passivated in accordance with QQ-P-35.

3.4.1 Heat treat. Heat treatment of corrosion-resistant steel shall be in accordance with MIL-H-6875. Heat treatment of titanium shall be in accordance with MIL-H-81200. Heat treatment of nickel alloy 718 shall be in accordance with AMS5662.

3.5 Design and construction. The design and construction of the bolt shall be as specified herein and on the applicable MS sheets.

3.5.1 Visual and dimensional requirements. Bolts manufactured to this specification shall meet the visual and dimensional requirements when tested in accordance with 4.5.1.

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3.5.2 Shank straightness. The shank straightness shall be within the values specified in table I when measured as specified in 4.5.2.

TABLE I. Shank straightness.

Shank diameter (inch)	Maximum deviation of bolt shank from surface plate (inch per inch grip length)
0.1900	0.0040
0.2500, 0.3125	0.0030
0.3750, 0.4375	0.0025
0.5000 and up	0.0020

3.5.3 Locking device. All bolts shall be of such design as to be retained in position by the locking balls or elements until the release mechanism is manually actuated.

3.5.3.1 Locking device actuating plunger. An axial pushing force shall release the locking mechanism. The force shall be applied to the actuating plunger located either in the center of the bolt head or the center of the thread end of the bolt. The locking mechanism shall automatically return to the locked position when the actuating force is removed.

3.5.3.2 Lubrication. Plungers and locking element, with the exception of the balls, shall be lubricated with solid film lubricant in accordance with MIL-L-46010. For plungers requiring passivation, lubrication shall be subsequent to passivation.

3.5.4 Release mechanism. The release mechanism shall be of such design as to remain in the locked position unless actuating force is applied.

3.5.4.1 Release mechanism actuating force. The actuating force required to release the locking device shall be within the range specified for the applicable size bolt shown in table II.

3.5.4.2 Endurance. The release mechanism shall, after 5000 cycles of operation, meet the values specified in table II (see 4.5.4.1).

3.5.5 Locking ball or element push-out force. Balls or elements used as a locking device shall be so retained in the shank as to withstand the minimum push-out force specified in table II (see 4.5.5). This requirement does not apply to bolts with locking elements that are fully retained by the plungers.

3.5.6 Operability. The release mechanism shall be operable when the bolt is loaded in shear to the following percentages of the minimum shear strength as shown in table II (see 4.5.6).

a. 0.1900 shank diameter - 35 percent.

b. 0.2500 shank diameter and larger - 50 percent.

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TABLE II. Performance characteristics.

Nominal size	Body double shear strength pounds (minimum)			Bolt tensile strength pounds (minimum) 2/	Retaining device tensile strength pounds (minimum)	Release mechanism actuating force lb		Bolt removal force lb max	Locking balls push-out force lb min
	A286 1/	A286 Ti	PH13-8Mo 718	PH13-8Mo, 718 A286 and Ti	PH13-8Mo, 718 A286 and Ti	Min	Max		
.1900	5,150	4,250	5,150	1,110	110	0.5	5	6	5
.2500	9,200	7,360	9,200	2,040	230	1	9	10	10
.3125	14,400	11,500	14,400	3,250	510	2	9	10	10
.3750	20,600	16,500	20,600	5,050	575	2	10	11	10
.4375	28,200	22,500	28,200	6,800	710	3	11	12	20
.5000	36,800	29,400	36,800	9,250	1160	3	12	13	20
.5625	46,500	37,400	46,500	11,800	1420	3	12	13	20
.6250	58,600	44,800	58,600	14,400	2070	4	12	13	25
.7500	82,400	62,800	82,400	20,900	2950	5	14	15	30
.8750	117,000	89,400	117,000	28,600	3980	5	14	15	30
1.0000	154,000	113,000	154,000	37,600	5200	5	14	15	30

1/ Inactive for new design after 31 December 1990.
2/ Tensile loads are based on the strength of the companion nut: AN310, AN320, MS17825, MS17826, MS21224 and MS21244.

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3.5.6.1 Locking ball or element retraction. For head actuated bolts, the locking ball or element shall retract or release when the release mechanism actuating force is applied. This release allows removal of the bolt from a bushing when the bolt removal force is applied along the bolt axis opposing the release mechanism actuating force (see table II). For thread end actuated bolts, the locking ball or element shall retract or release when the bolt removal force of table II is applied to the release mechanism. This release allows removal of the bolt from a bushing when the release mechanism force is applied along the bolt axis in the same direction as the bolt removal force (see 4.5.6.1).

3.5.7 Cotter pin hole. A cotter pin hole shall be drilled in accordance with the applicable MS sheets. This hole shall not interfere with the operation of the retaining mechanism, and all burrs shall be removed. Countersinking of cotter pin hole is optional.

3.5.8 Double shear strength. The bolt shall conform to the minimum double shear strength as specified in table II (see 4.5.7).

3.5.9 Retaining device tensile loads. The retaining device tensile loads shall be as specified in table II (see 4.5.8).

3.5.10 Bolt tensile test. The bolt tensile loads shall be as specified in table II (see 4.5.9).

3.5.11 Surface roughness. The surface roughness of the bolts shall not exceed the values shown in table III (see 4.5.10).

TABLE III. Roughness height rating.

Area	Maximum surface roughness (microinches)
Shank and underside of head	63
Other surfaces	125

3.5.12 Threads.

3.5.12.1 Form and dimensions. Unless otherwise specified, the threads shall be right-handed and the dimensions, form and contour shall conform to MIL-S-8879 and the applicable MS sheets.

3.5.12.2 Rolling. Threads shall be fully formed by a single rolling process.

3.5.12.3 Runout. Thread runout, consisting of a maximum of 1.500 incomplete thread pitches, shall fair onto the shank, thereby eliminating any abrupt changes in cross-sectional area. There shall be a minimum thread runout of 0.5 pitch. Bottom and sides of runout thread pitches may deviate from true thread form but shall be smooth.

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3.5.12.4 Concentricity of shank to threads. The concentricity of shank diameter "D" and thread pitch diameters shall be within "Y" values Full Indicator Movement (FIM) (see figure 1).

3.5.13 Head. The bolt heads shall be upset or forged.

3.5.13.1 Bearing surface. The bearing surface of bolt protruding heads shall be at right angles to the shank within the limits shown on figure 1. The angular variations of the underside of the head shall be uniform around the shank within a tolerance of +10 minutes, as measured between the bearing surface of the head and shank, at a length along the shank from the head equal to the diameter of the bolt (see 4.5.1).

3.5.13.2 Concentricity of head to shank. For flush-headed fasteners, the conical surface of the head and the shank diameter "D" shall be within 0.005 FIM. Head driving recess to the shank shall be within 0.016 FIM for sizes 0.2500 and smaller and within 0.024 FIM above 0.2500. Concentricity of the head and shank of the protruding headed fasteners shall be within "X" values FIM (see figure 1).

3.5.14 Environmental limits.

3.5.14.1 Temperature. Bolts shall function satisfactorily and be operable at temperatures from -65° to +450°F for bolts fabricated from AMS4928 or AMS4967 (6Al-4V) and from AMS5629 (PH13-8Mo) and at temperatures from -65° to +650°F for bolts fabricated from AMS5737 (A286) and AMS5662 (718) or AMS5663. Bolts shall be tested as specified in 4.5.13.

3.5.14.2 Dust (fine sand). Bolts shall be tested as specified in 4.5.14.

3.5.14.3 Corrosion. Bolts shall be tested as specified in 4.5.15.

3.5.15 Metallurgical examination. Flow lines shall be in accordance with figure 2. Bolts shall be tested as specified in 4.5.16.

3.5.15.1 Discontinuities. Each bolt shall be examined for discontinuities as specified in 4.5.17, 4.5.17.1 and 4.5.17.2. Any bolt having discontinuities exceeding the limitations in 3.5.15.1.1, 3.5.15.1.2 and 3.5.15.1.3 shall be rejected.

3.5.15.1.1 Cracks. Bolts shall be free of cracks in any direction or location.

3.5.15.1.2 Laps, seams and inclusions. Finished bolts shall not possess laps, seams or inclusions that exceed the depth shown in table IV.

3.5.15.1.3 Thread discontinuities. Threads shall have no laps or seams below the pitch diameter (see figure 4). Laps and seams that are above the pitch diameter (see figure 3) and do not exceed the depth limits of table IV are permissible.

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TABLE IV. Metallurgical depths.

Bolt diameter (inch)	Depth of laps, seams and inclusions (inch, max)	Bolt head top decarburization or carburization (inch, max) (see 3.5.15.3.3)	Bolt head top titanium overheating (inch, max) (see 3.5.15.3.1)
0.1900	0.003	0.003	0.010
0.2500	0.004	0.003	0.015
0.3125	0.005	0.003	0.015
0.3750	0.006	0.003	0.020
0.4375	0.007	0.003	0.020
0.5000	0.008	0.004	0.020
0.5625	0.008	0.004	0.020
0.6250	0.008	0.004	0.020
0.7500	0.008	0.004	0.020
0.8750	0.008	0.004	0.020
1.0000	0.008	0.004	0.020

3.5.15.2 Hydrogen content. Bolts manufactured from 6Al-4V titanium alloy shall not exhibit hydrogen content in excess of 0.0125 percent (125 PPM). Samples shall be examined in accordance with 4.5.18.

3.5.15.3 Surface effects and microstructure.

3.5.15.3.1 Class 3. Bolts manufactured from 6Al-4V titanium alloy shall be free of bursts, voids or gross alloy segregation.

3.5.15.3.2 Overheating (class 3 only). Microstructure shall be free from overheating, that is, the material shall be free from indications that it has been heated to a temperature above Beta Transus without subsequently receiving significant mechanical reduction in the Alpha-Beta temperature range. Structure which has outlines of equiaxial prior all-Beta grain and no primary Alpha is considered overheated. Slight overheating adjacent the top surface of the head is permissible within the depth limits specified in table IV.

Parts shall exhibit no surface contamination on the bearing surface of the head, head-to-shank fillet, shank and threads. Nonbearing surfaces of the head may show contamination not exceeding 0.003-inch in depth, when tested in accordance with 4.5.18 and 4.5.19.

3.5.15.3.3 Carburization and nitriding (class 1 only). There shall be no evidence of carburization or nitriding on the bearing surface of the head, head-to-shank fillet, shank or threads when examined in accordance with 4.5.20.

3.6 Performance. The bolt shall pass the performance characteristics of table II when subjected to the applicable tests of Section 4.

3.6.1 Vibration. Bolts shall withstand the vibration test specified in 4.5.21.

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3.6.2 Stress durability. Except for AMS5662 and AMS5663 (718), bolts shall withstand the specified load for the specified time without failing or developing cracks (see 4.5.22).

3.6.3 Ductility. For bolts manufactured from AMS5662 or AMS5663 (718) material ductility shall be not less than 8 percent elongation and 15 percent reduction in area when tested as specified in ASTM E8 (see 4.5.23).

3.7 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable.

3.8 Identification of product. Each bolt shall be marked in accordance with the applicable standard on top of the head or side of the flange. The manufacturer's identification and code marking for traceability by production lot shall be marked on the top or side of the head. Marking may be formed by forging, stamping or laser. Marking may be raised or indented 0.010 inch maximum.

3.9 Workmanship. Workmanship shall be consistent with the type of product, finish and class of thread fit. Sharp edges shall be broken; hanging burrs and slivers which might become dislodged during use shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall consist of all the examinations and tests listed in table V.

4.3.1 Sample size. The qualification samples for bolt sizes 0.1900 through 0.5000-inch diameter shall consist of 28 bolts for each size for which qualification is desired. Sample bolts shall have a minimum grip length of 1.3125 inches. The qualification samples for bolt sizes 0.5625-inch and larger shall consist of 20 bolts for each size for which qualification is desired. The sample bolts shall have a minimum grip length of 4D. In addition, the manufacturer shall supply 28 nuts in accordance with MS21133 or equivalent and 28 washers (plain) in accordance with MS20002 for sizes 0.1900

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TABLE V. Qualification inspection.

Examinations or tests	Number of samples to be tested	Requirement paragraph	Test method paragraph
Visual and dimensional requirements	3	3.5.1	4.5.1
Shank straightness	3	3.5.2	4.5.2
Locking device	3	3.5.3	4.5.3
Release mechanism	3	3.5.4	4.5.4
Endurance	3	3.5.4.2	4.5.4.1
Locking ball push-out force	3	3.5.5	4.5.5
Operability	3	3.5.6	4.5.6
Locking ball or element retraction	3	3.5.6.1	4.5.6.1
Double shear strength	3	3.5.8	4.5.7
Retaining device tensile loads	3	3.5.9	4.5.8
Bolt tensile test	3	3.5.10	4.5.9
Surface roughness	3	3.5.11	4.5.10
Threads	3	3.5.12	4.5.11
Concentricity of shank to threads	3	3.5.12.4	4.5.12
Concentricity of head to shank	3	3.5.13.2	4.5.12
Temperature	3	3.5.14.1	4.5.13
Dust (fine sand)	3	3.5.14.2	4.5.14
Corrosion	3	3.5.14.3	4.5.15
Metallurgical examination	1	3.5.15	4.5.16
Discontinuities	3	3.5.15.1	4.5.17
Hydrogen content (Class 3 only)	1	3.5.15.2	4.5.18
Overheating (Class 3 only)	1	3.5.15.3.2	4.5.19
Carburization and nitriding (Class 1 only)	1	3.5.15.3.3	4.5.20
Vibration	5	3.6.1	4.5.21
Stress durability (except Class 4)	3	3.6.2	4.5.22
Ductility (Class 4 only)	1	3.6.3	4.5.23

through 0.5000-inch, and 20 nuts in accordance with MS21133 (except non-self-locking) or equivalent in strength and 28 washers (plain) in accordance with MS20002 for sizes 0.5625-inch and larger. Samples shall be identified and forwarded to the activity responsible for qualification (see 6.3).

4.3.2 Qualification test report. The manufacturer shall furnish a certified test report showing that the manufacturer's product conforms to the requirements of this specification. The test report shall include, as a minimum, actual results of the tests specified herein. When this report is submitted, it shall be accompanied by a dated drawing which describes the manufacturer's product by specifying all dimensions and tolerances, composition of material selected, finish applied and the heat treatment. The manufacturer's part number for each size and length shall be included on the dated drawing.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in table VI.

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TABLE VI. Quality conformance inspection.

Type of test	Sampling plan	Requirement paragraphs	Test paragraphs
Visual and dimensional requirements	P	3.5.1	4.5.1
Shank straightness	P	3.5.2	4.5.2
Locking device	R	3.5.3	4.5.3
Release mechanism	Q	3.5.4	4.5.4
Endurance	Table VIa	3.5.4.2	4.5.4.1
Locking ball push-out force	100%	3.5.5	4.5.5
Double shear strength	Table VIa	3.5.8	4.5.7
Retaining device tensile loads	Table VIa	3.5.9	4.5.8
Bolt tensile test	Table VIa	3.5.10	4.5.9
Surface roughness	P	3.5.11	4.5.10
Metallurgical examination	R	3.5.15	4.5.16
Discontinuities	100%	3.5.15.1	4.5.17
Hydrogen content (Class 3 only)	R	3.5.15.2	4.5.18
Overheating (Class 3 only)	R	3.5.15.3.2	4.5.19
Carburization and nitriding (Class 1 only)	R	3.5.15.3.3	4.5.20
Stress durability (except Class 4)	R	3.6.2	4.5.22
Ductility (Class 4 only)	R	3.6.3	4.5.23

TABLE VIa.

Inspection Level S-3	Sample Size Code Letter (From MIL-STD-105, Table II-C)								
	A	B	C	D	E	F	G	H	J
Sample Size	2	2	2	3	5	8	13	20	32
Percent Defective	0	0	0	0	0	0	0	0	0

Sampling Plan

P. MIL-STD-105, Inspection Level S-3, AQL 1.5 percent defective.
Q. MIL-STD-105, Inspection Level S-3, AQL 1.0 percent defective.
R. As specified in Test paragraph, failure shall be cause for rejection.

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4.4.1 Sampling plan. Statistically, sampling and inspection shall be in accordance with MIL-STD-105 and table VI. Samples for destructive tests may be selected from those that have been subjected to and passed non-destructive tests with additional units selected at random from the inspection lot.

4.4.1.1 Classification of defects. Classification of defects shall be in accordance with table VII. Defects shall be classified as major or minor in accordance with MIL-STD-105. The AQLs for major and minor defects shall be as follows:

Major defects	1.5 percent
Minor defects	4.0 percent

TABLE VII. Classification of defects.

Category	Defect
Major	
101	Thread dimensions and form
102	Shank
103	Grip length
104	Fillet from head to shank
105	Incomplete threads
106	Cotter pin hole location
107	Hex dimension across flats
108	Shank straightness
109	Squareness of bearing surface to shank
110	Concentricity of shank to threads
111	Burrs in cotter pin hole and on bearing surface under head
112	Slot in head
113	Protrusion of locking balls or elements
114	Material identification
115	Concentricity of head and shank
Minor	
201	Length
202	Head thickness
203	Length at hex corner
204	Cotter pin hole diameter
205	Bearing surface under head
206	Burrs (general)
207	Slot dimensions and location
208	Surface roughness
209	Chamfer on thread end
210	Radial location of locking balls
211	Hex dimensions across corners
212	Chamfer on rounding bearing surface of head

4.4.2 Inspection lot. An inspection lot shall consist of sample bolts selected at random from one production lot per the applicable AQL.

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4.4.2.1 Production lot. A production lot shall consist of finished bolts that are of the same head configuration, shank diameter, grip length, thread size and all other requirements, fabricated from the same material and heat lot processed at the same time and submitted to inspection at the same time.

4.4.3 Rejected lots. When an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units and resubmit for reinspection. Resubmitted lots shall be tested using tightened inspection. Where the original acceptance number was zero, a sample size represented by the next higher sample size code letter shall be selected. When inspection lots are resubmitted, full particulars concerning the cause of previous rejection and the action taken to correct defects shall be furnished by the contractor to the acquiring activity. Resubmitted lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4.4 Inspection reports. The supplier shall furnish a certified report with each shipment of bolts, stating the quantitative results of tests and inspection performed on representative samples. This report shall include the necessary information to correlate the test report with the purchase order or contract.

4.5 Test methods.

4.5.1 Visual and dimensional requirements. Samples selected in accordance with 4.4.1 shall be examined for conformance to the requirements listed in table VII and on the applicable standard.

4.5.2 Shank straightness. Shank straightness shall be determined by rolling the shank of the bolt with the locking element retracted on a surface plate and measuring the point of greatest deviation with a 0.5000-inch wide feeler gage.

4.5.3 Locking device. All bolts in a lot shall be tested for protrusion of locking balls or elements and operation of release mechanism.

4.5.4 Release mechanism. Sample bolts shall be mounted in a fixture conforming to figure 5 and the release actuating force measured for conformance to the values in table II.

4.5.4.1 Endurance. The release mechanism shall endure 5000 cycles of operation as specified in 4.5.4. One cycle shall be defined as release of the locking device followed by return to the locked position (see figure 5). At the end of the cycles, the actuating force required to release the locking device shall be measured for conformance to table II.

4.5.5 Locking ball push-out force. The plunger and one ball shall be removed from the test sample. The sample shall then be placed on a V-shaped anvil, with the remaining ball in the hole placed downward, in a manner not permitting contact with any portion of the anvil. A mandrel connected to a direct reading load indicator shall be inserted through the empty top ball hole, and the loads specified in table II applied against the ball in the bottom staked hole.

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4.5.5.1 Locking elements. If the locking element is retained by the plunger, no push-out test is required, but they must pass the vibration test during qualification.

4.5.6 Operability. The shear load specified in 3.5.6 for the applicable size bolt shall be applied for one minute and then removed. The release actuating force shall then be measured for conformance to the values in table II. The test shall not be conducted on bolts with a grip length less than 2D.

4.5.6.1 Locking ball or element retraction. Bolts used for the test specified in 4.5.6 shall be installed in a fixed steel bushing with the hole being in accordance with NAS 618, Column "C." The maximum release mechanism actuating force specified in table II shall be applied to the locking device actuating button. The maximum bolt removal force specified in table II shall be applied along the bolt axis. The bolt shall pass completely out of the bushing hole.

4.5.7 Double shear strength. Bolts with a grip length of 2D to 4.5D for a protruding head and 2.5D to 4.5D for a flush head shall be tested adjacent to the head. For bolts with a grip length less than 2D for a protruding head and 2.5D for a flush head, one bolt shall be prepared greater than 2D or 2.5D, respectively, from the same production lot, except for grip length, and a double shear test shall be performed. The radius under the bolt head can be removed by machining if it interferes with the shear test. Bolts with a grip length greater than 4.5D for the protruding head and 5D for the flush head up to 6.5D shall be tested adjacent to the head and within 0.5D of the retaining device. Bolts with a grip greater than 6.5D for the protruding head and 7D for the flush head shall be tested in three separate locations:

a. Within 0.5D of the retaining device.

b. Adjacent to the head.

c. In the center of the grip.

4.5.8 Retaining device tensile loads. The retaining device tensile loads specified in table II shall be applied to the bolt (without a nut) in accordance with MIL-STD-1312, Test 8. The retaining device shall be seated on a bushing, heat treated to not less than Rc63. The hole in the bushing shall be in accordance with Test 8. Bolts with a grip length less than 2D need not be tested.

4.5.9 Bolt tensile test. Sample bolts shall be tested in tension between the head of the bolt and a mated nut which will ensure failure of the bolt. The test shall be performed in accordance with MIL-STD-1312, Test 8. A spacer or washer that clears the ball shall be used to prevent damage to the fixture. Bolts with a grip length less than 2D need not be tested.

4.5.10 Surface roughness. Surface roughness shall be measured in accordance with ANSI B46.1.

4.5.11 Threads. The threads shall be examined for conformance to 3.5.12.

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4.5.12 Concentricity. Concentricity of head to shank and shank to threads shall be measured by observing the total variation of a dial indicator testing the head or threads (as applicable) as the bolt is rotated with its shank as an axis.

4.5.13 Temperature.

a. For Class 1, 2, 3 and 4 bolts

Minimum temperature: Sample bolts shall be held at the minimum temperature of -65 +10°F for 1 hour, and then tested at room temperature for conformance to 3.5.4 and 3.5.4.1.

b. For Class 1 and Class 3 bolts

Maximum temperature: Sample bolts shall be held at the maximum temperature of 450 +10°F for 1 hour, and then tested at room temperature for conformance to 3.5.4 and 3.5.4.1.

c. For Class 2 and 4 bolts

Maximum temperature: Sample bolts shall be held at the maximum temperature of 650 +10°F for 1 hour, and then tested at room temperature for conformance to 3.5.4 and 3.5.4.1.

4.5.14 Dust (fine sand). Sample bolts shall be subjected to the dust test of MIL-STD-810, Method 510.2. After completion of the test, the release mechanism shall operate within limits specified in this specification.

4.5.15 Corrosion. Sample bolts shall be subjected to the salt spray test of MIL-STD-1312, Test 1. After completion of the test, the release mechanism shall operate within limits specified in this specification. Small amounts of rust discoloration on the balls manufactured from QQ-S-763 class 440C is permissible.

4.5.16 Metallurgical examination. One bolt shall be selected at random regardless of lot size for grain flow examination. Head structure and grain flow of fillet radius and threads shall be taken by macroexamination at a magnification of 10X. Flow lines shall be indicative of a forged product (see figure 2) following the general contour of the head. They shall be continuous, except that slight cutting of the flow lines in the head to shank fillet, due to the oxide removal process, shall be acceptable. Flow lines are more difficult to resolve in CRES and super alloys (with respect to alloy steel) due to their cleanliness and homogeneity. It is recommended that flow line characterization be performed in the as-forged condition and as shown in figure 6. For fully heat treated parts, etching times may need to be increased substantially.

4.5.17 Discontinuities. Fluorescent penetrant or magnetic particle inspection shall be used to determine the presence of cracks and discontinuities such as laps, seams and inclusions. Fluorescent penetrant or magnetic particle indications of themselves shall not be cause for rejection. If indications are considered cause for rejection, representative samples

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shall be further examined by microexamination to determine if the indicated discontinuities exceed the limits specified in table IV.

4.5.17.1 Fluorescent penetrant inspection. Fluorescent penetrant inspection shall be performed in accordance with MIL-STD-6866 for AMS5737 (A286), AMS4928 (6Al-4V) or AMS4967 (6Al-4V) or AMS5662 (718) or AMS5663 (718). Only the bolt body shall be fluorescent penetrant inspected. Such inspection shall be performed on finished bolts, subsequent to any processing operations that could adversely affect the part. Bolt bodies shall not be marked for identification of fluorescent penetrant inspection.

4.5.17.2 Magnetic particle inspection. Magnetic particle inspection shall be performed in accordance with MIL-STD-1949 for AMS5629 (PH13-8Mo) parts. Only the bolt body shall be magnetic particle inspected. Such inspection shall be performed on the finished bolt bodies subsequent to any processing operations which could adversely affect the part. Bolt bodies shall not be marked for identification of magnetic particle inspection.

4.5.18 Hydrogen content (Class 3 only).

4.5.18.1 Sampling and acceptance criteria. One bolt shall be selected at random regardless of lot size. The test section shall meet the acceptance condition as specified in 3.5.15.2.

4.5.18.2 Procedure. A section of material shall be taken from the bolt in accordance with figure 6 and shall be subjected to a hydrogen determination analysis. Use procedures and equipment that are capable of analyzing hydrogen to an accuracy of +0.0010 percent (10 ppm). Plating and lubricant shall be removed. Mechanical removal is acceptable provided the depth of material removed does not exceed 0.003-inch.

4.5.19 Overheating (Class 3 only). One sample part shall be micro-examined at 100X or greater. Parts shall be free from overheating in accordance with the requirements of 3.5.15.3.1.

4.5.20 Carburization and nitriding (Class 1 only). One sample part shall be microexamined along longitudinal section at 100X or greater in accordance with figure 6. Examination shall reveal no carburization or nitriding in accordance with the requirements of 3.5.15.3.3.

4.5.21 Vibration. Sample bolts, 0.6250-inch diameter and smaller, shall be subjected to the vibration test in accordance with MIL-STD-1312, Test 7 and in accordance with table VIII. The bolts shall be installed without nuts or preload; a MS20002 countersunk washer may be used between the vibration fixture and the retaining device. Vibration life of less than 30,000 cycles shall be cause for rejection. Vibration tests on bolts larger than 0.6250-inch diameter are waived provided that the 0.6250-inch and smaller bolts of the same type and design have satisfactorily passed the vibration test.

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TABLE VIII. Hole size and grip lengths for vibration test.

Bolt size (in.)	C diameter (in.) 1/	Bolt grip length (in.)		Bolts required
		Hex head or pan head	Flush head 2/	
0.1900	0.1910/0.1930	1.312	1.375	5
0.2500	0.2510/0.2530	1.375	1.500	5
0.3125	0.3135/0.3155	1.375	1.500	5
0.3750	0.3760/0.3780	1.375	1.562	5
0.4375	0.4385/0.4405	1.438	1.625	5
0.5000	0.5010/0.5030	1.438	1.625	5
0.5625	0.5635/0.5655	1.500	1.688	5
0.6250	0.6260/0.6280	1.500	1.688	5

1/ "C" diameter is for new washer and cylinder ID from MIL-STD-1312, Test 7.
2/ Washer and cylinder chamfers shall not be increased for flush head bolts.

4.5.21.1 Vibration test failure. The bolts shall have failed the vibration test under the following conditions:

a. If any structural failure, such as broken plunger or spring, or locking element fall out, occurs during the test.

b. If any self-retaining bolt can be removed from the arbor and sleeve with the fingers, without pressing on the plunger.

c. If any self-retaining bolt cannot be removed from the arbor and sleeve with an applied release mechanism actuating force specified in table II.

4.5.22 Stress durability. The stress durability test shall be conducted on a minimum of one sample per lot using the sustained external load method in accordance with MIL-STD-1312, Test 5. The induced load shall be 75 percent of the tested ultimate tensile strength of the bolts. Free spinning, cadmium plated MS21133 or equivalent nuts shall be used with self-retaining bolts for this test. The bolts shall be in the stressed condition for 72 hours. The assemblies shall be examined each day for maintenance of load. The test shall be continued an additional day for each day that the examination was not performed. Nuts required for this test shall be new and shall not be used more than once in performance of this test.

4.5.23 Ductility. The specimen for the ductility test shall be prepared from a section of the raw material, processed with the production lot, or manufactured from the finished bolt. The ductility test shall be conducted on one specimen as specified in ASTM E8.

4.6 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with Section 5.

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5. PACKAGING

5.1 Preservation. Each bolt shall have the shank, retaining element and threads protected by a sleeve extending over the full length of the shank, retaining element and thread. The sleeve shall be manufactured from cardboard, asphalt impregnated chipboard, plastic sleeve covering, or a spiral wrap of kraft paper over chipboard. The sleeve shall be lined with material conforming to MIL-B-121.

5.2 Packing (level A or level C). Bolts shall be packed in accordance with PPP-H-1581, level A or level C (see 6.2).

5.3 Marking. Interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The self-retaining bolts are primarily intended for use in shear applications at dynamic points in flight control linkages, engine control mechanisms, fuel controls, bell crank systems and other dynamic joint assemblies of aircraft where self-locking nuts should not be used and where the failure of cotter pin or locking wire would affect safety of flight or endanger personnel. Self retaining bolts are intended to be used with the following nuts: AN310, AN320, MS17825, MS17826, MS21224 and MS21244.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Part number and class of the bolt desired.

d. Quantity.

e. Level of packing (see 5.2).

f. Production lot sampling for laboratory verification tests (see 4.4.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the applicable Qualified Products List QPL No. 23964 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose

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to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Attention: AIR-53033, Navy Department, Washington, DC 20361; however, information pertaining to qualification of products may be obtained from the Naval Air Development Center, Code 60133, Warminster, PA 18974.

6.3.1 Qualification tests. Qualification tests will be authorized only upon presentation of certified test reports indicating that the bolts conform to or will conform to this specification and the applicable MS sheet or a standard approved by the activity responsible for qualification.

6.4 Alloy steel. Alloy steel parts, previously specified in MIL-B-23964B, are no longer acceptable and have been replaced by corrosion-resistant steel parts.

6.5 Subject term (keyword) listing.

Bell crank systems
Dynamic joint assemblies
Engine control mechanisms
Flight control linkages
Fuel controls

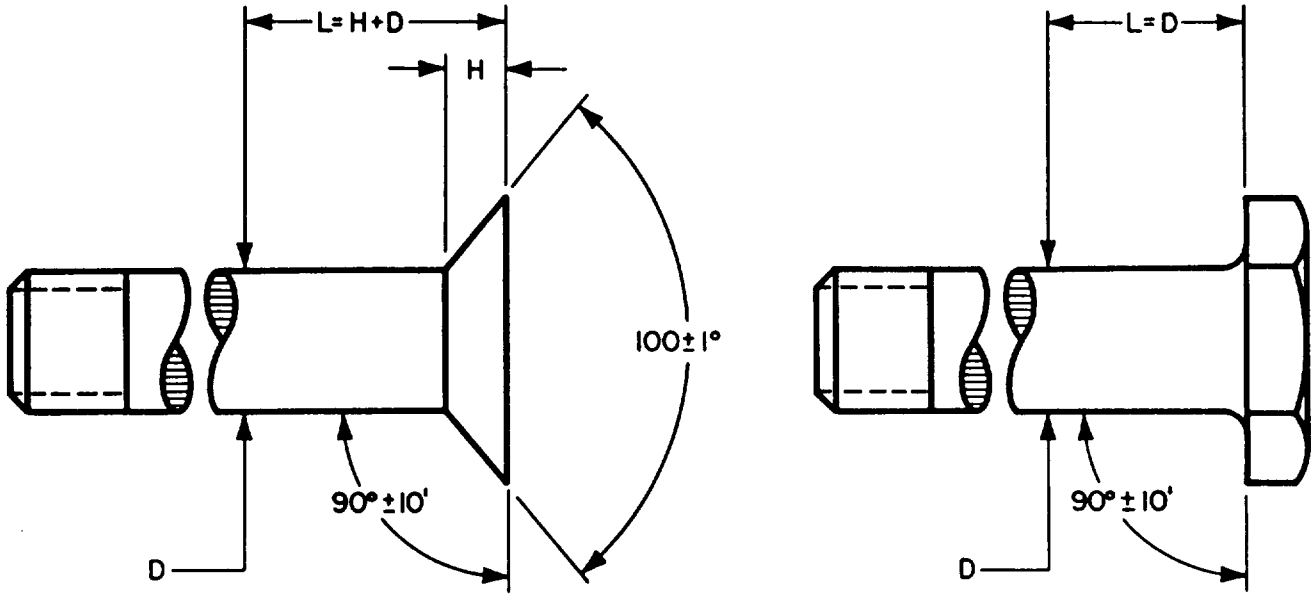
6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - AS
Air Force - 99

Review activities:
Air Force - 82

Preparing activity:
Navy - AS
(Project No. 5306-0763)

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Nom size	Y (see 3.5.12.4)	X (see 3.5.13.2)
0.1900	0.0045	0.005
0.2500	0.0045	0.006
0.3125	0.0045	0.008
0.3750	0.0060	0.009
0.4375	0.0060	0.010
0.5000	0.0060	0.011
0.5625	0.0060	0.012
0.6250	0.0060	0.015
0.7500	0.0090	0.018
0.8750	0.0090	0.020
1.0000	0.0090	0.022

FIGURE 1. Head angularity.

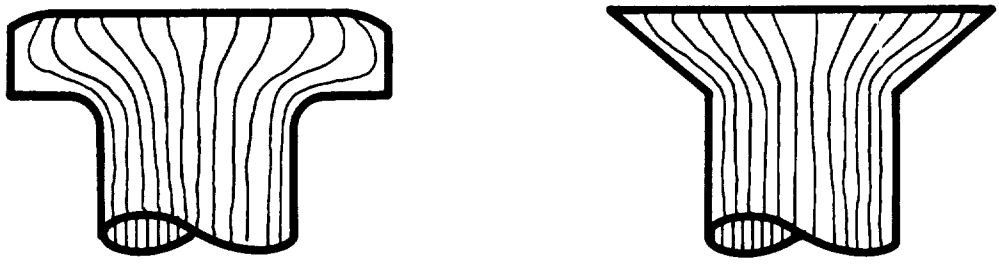


FIGURE 2. Grain flow in head.

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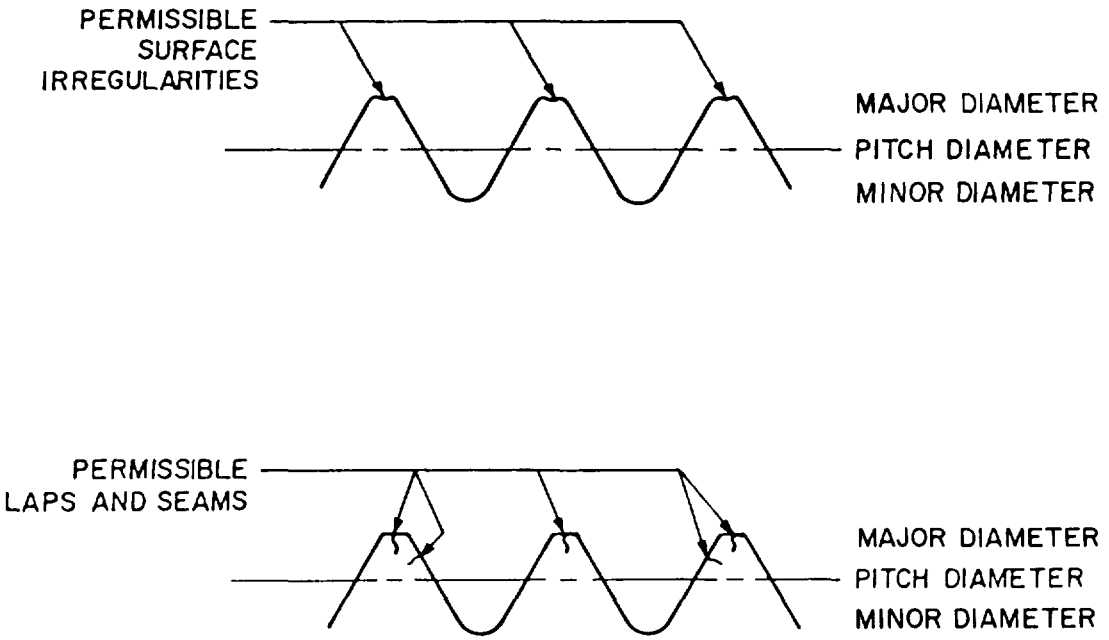


FIGURE 3. Permissible laps, seams and surface irregularities.

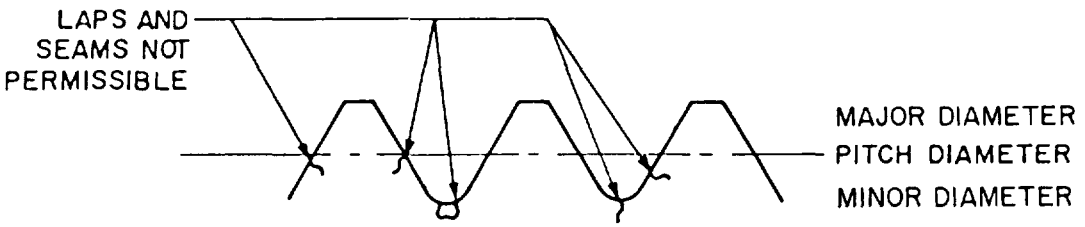


FIGURE 4. Nonpermissible laps, seams and surface irregularities.

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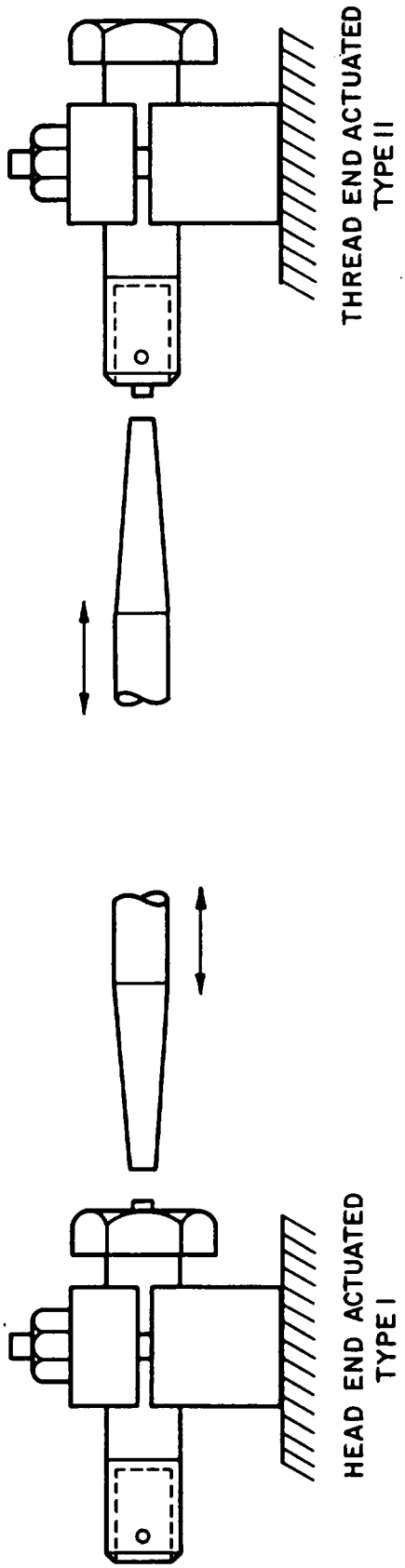
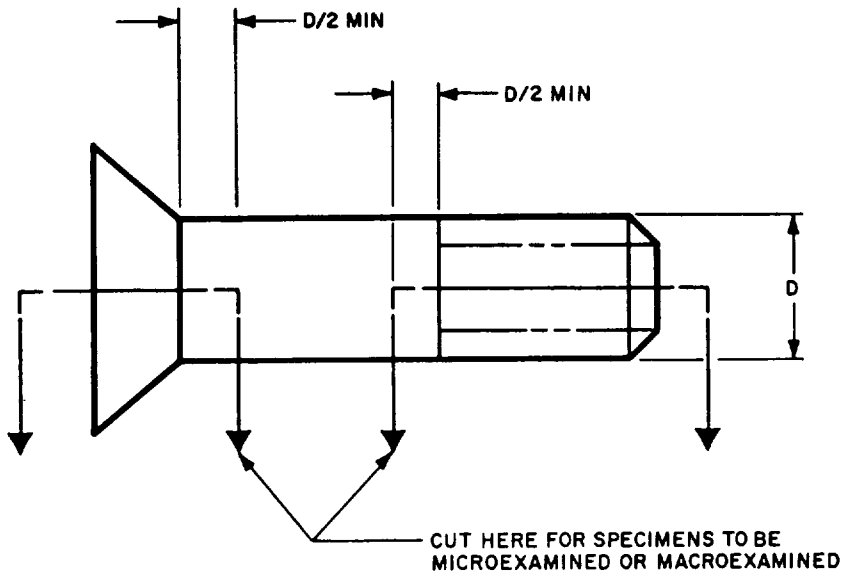


FIGURE 5. Stroke required to fully activate release mechanism.

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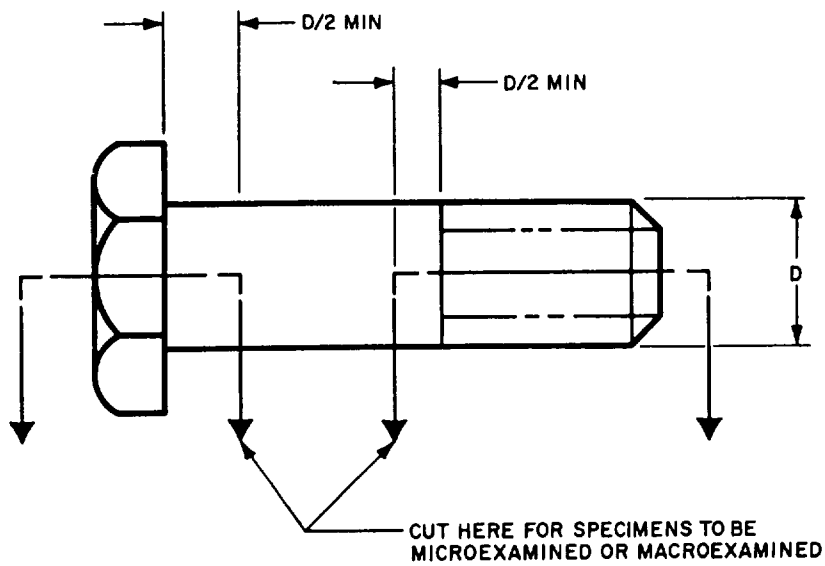


FIGURE 6. Metallurgical specimens.

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-23964D	2. DOCUMENT DATE (YYMMDD) 10 AUGUST 1990

3. DOCUMENT TITLE
BOLT, SELF-RETAINING, POSITIVE LOCKING

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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